Evolved Neural Agents as Tireless Playtesters

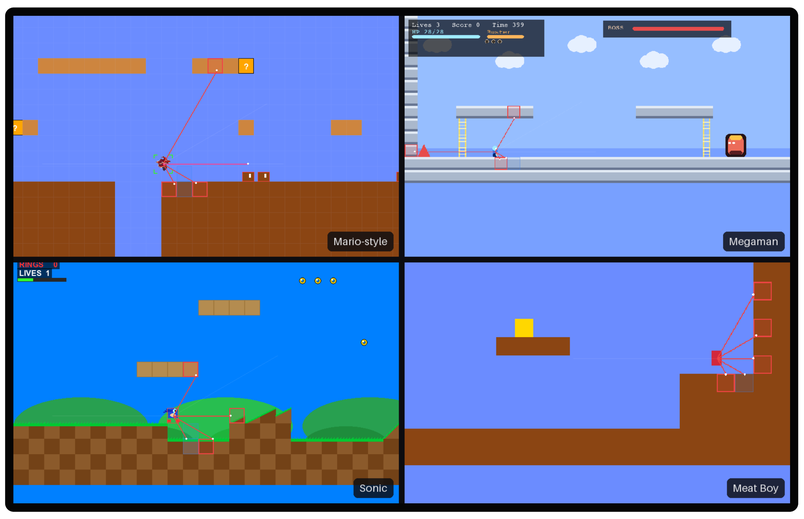
A deterministic, laptop-scale probe that measures game balance inside the level designer's edit loop

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Balance needs a thousand playthroughs. Humans give you ten.

- Level balance is tuned by playtesting: slow, costly, and **not repeatable**. Move one saw, and last week's testers cannot be re-run.
- Deep-RL test agents work in industry [1], but they need GPUs, reward shaping and per-game tuning: too expensive *per level edit*.
- Simple agents can map difficulty [2], and GAs rival deep RL [3]. Our GA probe reproduces the failure-mode taxonomy of our earlier PPO version at about **1/100 of the compute**.
- **Our goal: a regression test for game feel.** A frozen, cheap, deterministic playtester, so that a change in the metrics isolates the change in the level.

One probe, many engines

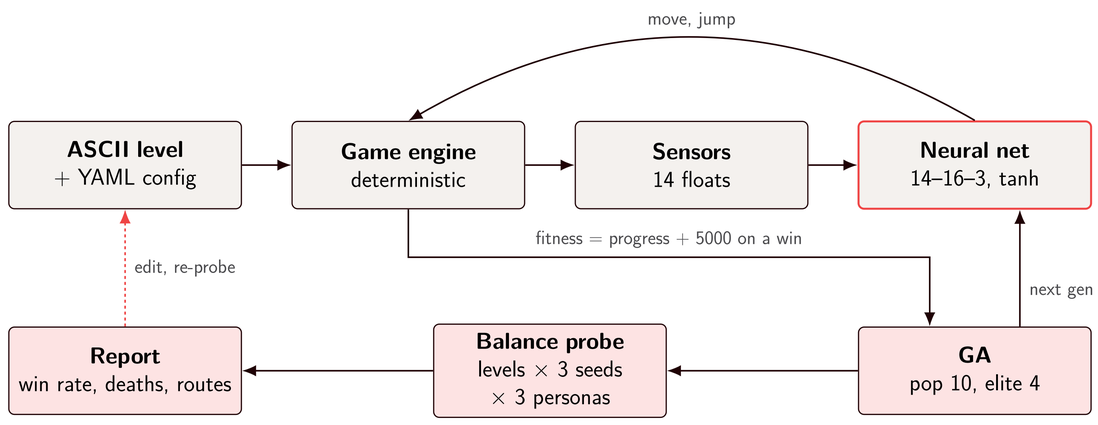


The red fan is **all the agent sees**: six raycasts, a pit probe, an enemy corridor, velocity and jump state.

14 numbers into a 291-weight network.

Four of the five hand-written engines are shown. The active roster is Mario-style, Meat Boy and a top-down Bomberman.

PEAK: play, measure, edit, repeat



- **Agent:** fixed-topology MLP, $14 \to 16 \to 3$ (**291 weights**). No gradients, no replay buffer, no reward shaping, no GPU.
- **Evolution:** population 10, elite 4, tournament k=5, uniform crossover 0.7, Gaussian mutation (rate 0.15, σ 0.15), annealed $\times$0.5 after a level's first win.
- **Probe:** a fresh population per level $\times$ seed; reports win rate $\pm$ 95 % CI, generations-to-first-win, death causes, death heat-map and routes.
- **One seed** drives initialisation, mutation and every game step: two runs are **bit-identical**.
- **Cost:** 5,000+ steps per second on a laptop. A Mario probe (2 levels $\times$ 3 seeds $\times$ 40 generations) takes about **six CPU-minutes**.

Personas: who is the level hard for?

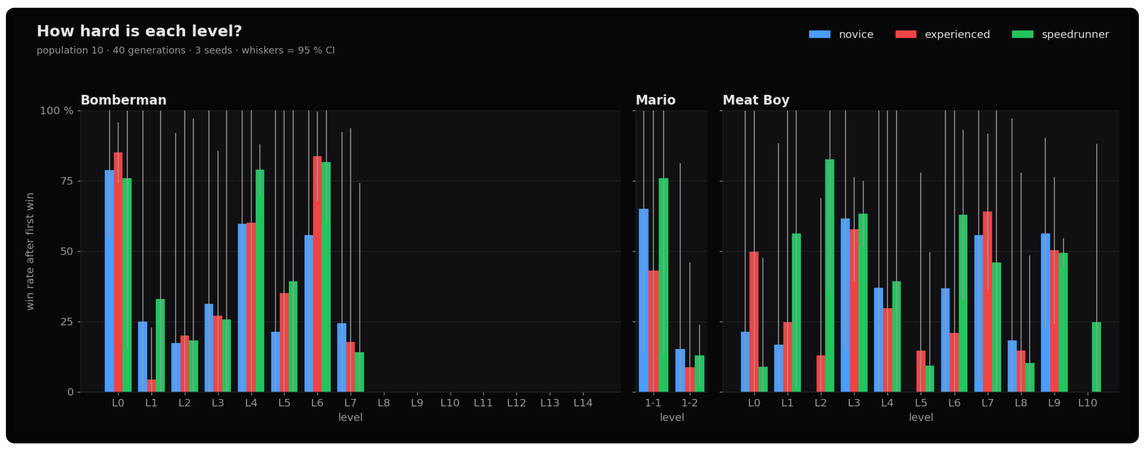
Persona	Capability	Objective
Novice	walk; senses refresh every 3rd frame	reach the goal
Experienced	walk; full reactions	reach the goal
Speedrunner	sprint; full reactions	goal + 25 fitness per second left

A persona is a capability + objective profile, not a reward function [4]. The win-rate gap between personas on one level is its skill-expression signal.

The probe in the design loop: Bomberman

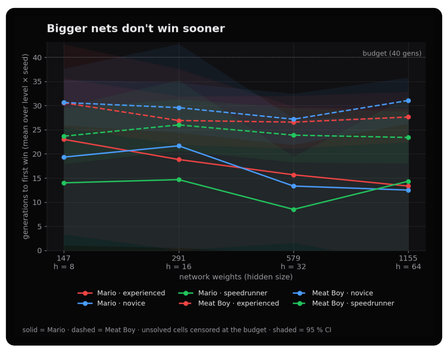
- A draft level was reported unsolvable, one rung after a trivial one: its bricks **sealed every route**. A reachability test now pins the intent of each level.
- The dominant death on every level was the agent's own bomb (55–97 %). Fuse 2.5 s $\to$ 4 s: level 6 went from 0/3 to 2/3 seeds.
- A range-1 bomb killed a wandering enemy in **0 of 200** trials. Enemy speed $\times$0.7 gave level 9 its first wins (0/3 $\to$ 2/3).
- Every change was A/B-probed on identical seeds: measured, not guessed.

Result 1: levels rank themselves

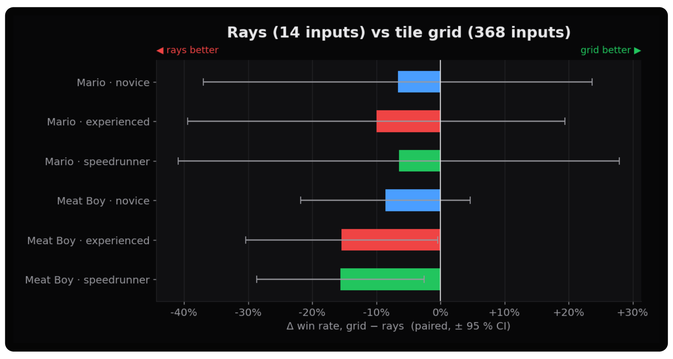


- **Mario:** 1-1 is solved by every seed and persona; 1-2 falls to a 9–15 % win rate.
- **Meat Boy:** from "every seed, every persona" (L3, L9) to a level that only the speedrunner solves (L10). The sprinting persona is the worst on L0 and the best on L1 and L2.
- **Bomberman:** a smooth ramp through L7, then **a cliff**. That step is what the tool exists to locate.

Result 2: the agent is not the bottleneck

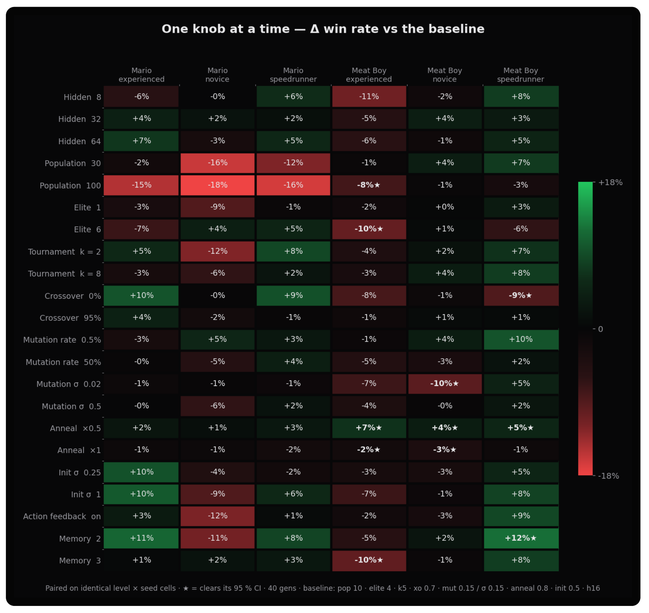


Bigger nets do not win sooner. Hidden size 8 to 64 (147 to 1,155 weights): generations-to-first-win stays inside overlapping 95 % CIs in all six sweeps.



Fourteen rays beat 368 grid cells in 6 of 6 sweeps (2 clear the 95 % CI). More input is more weights to evolve, not more insight.

Result 3: one knob at a time



- 22 variants $\times$ 6 sweeps. Each knob moves to its literature bound [5, 6], paired on identical level $\times$ seed cells.
- **One change wins everywhere:** anneal mutation $\times$0.5 after the first win. It is now the default.
- Population 100 is worse in 6 of 6, at ten times the compute.
- Elite 6, σ 0.02 and no crossover each hurt a Meat Boy persona significantly.
- Winners go to `ga_best.yaml` by strict majority vote.

Limitations and next steps

- Reactive agents are a *lower bound* on difficulty, not a model of a human. The probe is not yet validated against human play data.
- One-factor-at-a-time sweeps ignore interactions; three seeds give wide CIs on hard levels.
- Next: path and strategy-diversity metrics from logged trajectories, and risk/reward metrics from forced-collector agents.

References

[1] J. Bergdahl et al., "Augmenting Automated Game Testing with Deep Reinforcement Learning," IEEE CoG, 2020.
[2] A. Isaksen, D. Gopstein, A. Nealen, "Exploring Game Space Using Survival Analysis," FDG, 2015.
[3] F. P. Such et al., "Deep Neuroevolution: Genetic Algorithms Are a Competitive Alternative for Training Deep Neural Networks for RL," arXiv:1712.06567, 2017.
[4] C. Holmgård et al., "Evolving Personas for Player Decision Modeling," IEEE CIG, 2014.
[5] K. A. De Jong, "An Analysis of the Behavior of a Class of Genetic Adaptive Systems," PhD thesis, Univ. of Michigan, 1975.
[6] J. J. Grefenstette, "Optimization of Control Parameters for Genetic Algorithms," IEEE Trans. SMC 16(1), 1986.

Python + Pygame + numpy, built from scratch. 76 tests. No GPU.